Salman Sir's Structured Questions

S O U N D

1 Mark Question

1. Two sounds of same loudness and same pitch produced by two different instruments differ in their
(a) amplitudes
(b) frequencies
(c) waveforms
(d) All of these

2. The unit of quantity on which pitch of the sound depends is
(a) hertz
(b) metre
(c) metre/second
(d) second

3. Pitch of high frequency sound is
(a) low
(b) zero
(c) infinite
(d) high

4. By reducing the amplitude of a sound wave, its
(a) pitch decreases
(b) loudness decreases
(c) pitch increases
(d) loudness increases

5. The minimum distance between the source and the obstacle for an echo to take place is (speed of sound is 340 m/s)
(a) 172 m
(b) 18 m
(c) 17 m
(d) 34 m

2 Marks Questions

6. Distinguish between transverse wave and longitudinal wave.

7. Define echo and state its uses.

8. State the conditions for the formation of echo.

9. Define the term intensity of a sound wave. State the unit in which it is measured.

Structured Questions

10. (i) (a) Name the waves used for echo depth sounding.

(b) Give one reason of their use for the above purpose.
(c) Why are the waves mentioned by you not audible to us?
(ii) Differentiate between forced vibrations and resonance.
(iii) A person standing between two vertical cliffs and 640 m away from the nearest cliff shouted. He heard the first echo after 4 s and the second echo after 3 s later. Calculate
(a) the velocity of sound in air.
(b) the distance between the cliffs.

11. (i) Name the factor that determines
(a) loudness of the sound heard.
(b) quality of the note.
(c) pitch of the note.
(ii) Explain why musical instruments like the guitar are provided with a hollow box?
(iii) A sound wave of wavelength 0.332 m has a time period of 10^{-3} s. If the time period is decreased to 10^{-4} s, calculate the wavelength and frequency of the new wave.